

LEWIS PRESS LIMITED Generic Name Sorafenib Product Reference No PLFT07-161.00	Customer		Leaflet Pharmacode 1777	Separation Colours		Finished Folded Size 41 x 92 mm Proof Approval DATE:
	PharOS MT			(FRONT)	(BACK)	
	End Customer			1 - Black U	1 - Black U	
	Intega MY			ONLY colours listed in this column will be printed		
	Strength			Leaflet Dimensions		
200 mg		PRS LFT201_400x540 mm				
Proof Date		PRF No		CP Ref No		
10-04-2024		PRF_L07374		1,015		
Proof Version						
V2						

1. NAME OF THE MEDICINAL PRODUCT
inFENIB (sorafenib 200mg Film Coated Tablets)

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Each film-coated tablet contains 200 mg of sorafenib (as tosylate). For excipients, see 6.1.

3. PHARMACEUTICAL FORM

The product is presented as red-brown, round, biconvex film-coated tablets debossed with “200” on one side and plain on the other side with a diameter of tablet 12.0 mm ± 5%.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Hepatocellular carcinoma

inFENIB is indicated for the treatment of hepatocellular carcinoma.

Renal cell carcinoma

inFENIB is indicated for the treatment of patients with advanced renal cell carcinoma who have failed prior interferon-alpha or interleukin-2 based therapy or are considered unsuitable for such therapy.

Differentiated thyroid carcinoma

inFENIB is indicated for the treatment of patients with progressive, locally advanced or metastatic, differentiated (papillary/follicular/Hürthle cell) thyroid carcinoma, refractory to radioactive iodine.

4.2 Posology and method of administration

The recommended daily dose of sorafenib is 400 mg (2 x 200 mg tablets) taken twice a day, either without food or together with a low fat or moderate fat meal.

Method of administration

For oral use. To be swallowed with a glass of water.

Dosage regimen

Duration of treatment

Treatment should be continued until the patient is no longer clinically benefiting from therapy or until unacceptable toxicity occurs.

Dose titration, dose adjustment, special monitoring advice

Dose Reduction for Hepatocellular Carcinoma and advanced Renal Cell Carcinoma

Management of suspected adverse drug reactions may require temporary interruption and/or dose reduction of sorafenib therapy. When dose reduction is necessary during the treatment of hepatocellular carcinoma (HCC) and advanced renal cell carcinoma (RCC), the sorafenib dose should be reduced to two tablets of 200 mg once daily (see section Special warnings and precautions for use).

Suggested dose modifications for dermatologic toxicities are outlined in Table 1.

Table 1: Suggested Dose Modifications for Dermatologic Toxicities in Patients with Hepatocellular or Renal Cell Carcinoma

Dermatologic Toxicity Grade	Occurrence	Suggested Dose Modification
Grade 1: Numbness, dyesthesia, paresthesia, tingling, painless swelling, erythema or discomfort of the hands or feet which does not disrupt the patient's normal activities	Any occurrence	Continue treatment with inFENIB and consider topical therapy for symptomatic relief
Grade 2: Painful erythema and swelling of the hands or feet and/or discomfort affecting the patient's normal activities	1st occurrence	Continue treatment with inFENIB and consider topical therapy for symptomatic relief. If no improvement within 7 days, see below
	No improvement within 7 days or 2nd or 3rd occurrence	Interrupt inFENIB treatment until toxicity resolves to Grade 0–1. When resuming treatment, decrease inFENIB dose by one dose level (400 mg daily or 400 mg every other day)
	4th occurrence	Discontinue inFENIB treatment
Grade 3: Moist desquamation, ulceration, blistering or severe pain of the hands or feet, or severe discomfort that causes the patient to be unable to work or perform activities of daily living	1st or 2nd occurrence	Interrupt inFENIB treatment until toxicity resolves to Grade 0–1. When resuming treatment, decrease inFENIB dose by one dose level (400 mg daily or 400 mg every other day)
	3rd occurrence	Discontinue inFENIB treatment

Dose Reduction for Differentiated Thyroid Carcinoma

Management of suspected adverse drug reactions may require temporary interruption and/or dose reduction of sorafenib therapy.

When dose reduction is necessary during the treatment of differentiated thyroid carcinoma, the sorafenib dose should be reduced to 600mg daily in divided doses (two tablets of 200mg and one tablet of 200mg twelve hours apart).

If additional dose reduction is necessary, sorafenib may be reduced to one tablet of 200mg twice daily, followed by one tablet of 200mg once daily. After improvement of non-hematological adverse reactions, the dose of sorafenib may be increased.

Table 2: Recommended Dose Modifications for Dermatologic Toxicities for Patients with Differentiated Thyroid Carcinoma:

Dermatologic Toxicity Grade	Occurrence	inFENIB Dose Modification
Grade 1: Numbness, dyesthesia, paresthesia, tingling, painless swelling, erythema or discomfort of the hands or feet which does not disrupt the patient's normal activities	Any occurrence	Continue treatment with inFENIB
Grade 2: Painful erythema and swelling of the hands or feet and/or discomfort affecting the patient's normal activities	1st occurrence	Decrease inFENIB dose to 600 mg daily. If no improvement within 7 days, see below.
	No improvement within 7 days at reduced dose or 2nd occurrence	Interrupt inFENIB until resolved or improved to grade 1. If inFENIB is resumed, decrease dose. (See “Dose Reduction for Differentiated Thyroid Carcinoma”).
	3rd occurrence	Interrupt inFENIB until resolved or improved to grade 1. If inFENIB is resumed, decrease dose. (See “Dose Reduction for Differentiated Thyroid Carcinoma”).
	4th occurrence	Discontinue inFENIB permanently.

Grade 3: Moist desquamation, ulceration, blistering, or severe pain of the hands or feet, resulting in inability to work or perform activities of daily living.	1st occurrence	Interrupt inFENIB until resolved or improved to grade 1. If inFENIB is resumed, decrease dose by one dose level. (See “Dose Reduction for Differentiated Thyroid Carcinoma”).
	2nd occurrence	Interrupt inFENIB until resolved or improved to grade 1. When inFENIB is resumed, decrease dose by 2 dose levels. (See “Dose Reduction for Differentiated Thyroid Carcinoma”).
	3rd occurrence	Discontinue inFENIB permanently

Additional information on special populations

Pediatric Patients

The safety and effectiveness of sorafenib in pediatric patients has not been established.

Geriatric Patients

Elderly (above 65 years), Gender and Body Weight

No dose adjustment is required on the basis of patient age (above 65 years), gender, or body weight.

Patients with hepatic impairment

No dose adjustment is required in patients with Child-Pugh A or B hepatic impairment. Sorafenib has not been studied in patients with Child-Pugh C hepatic impairment (see section Pharmacokinetic Properties – Pharmacokinetics in special populations • Hepatic Impairment).

Patients with renal impairment

No dose adjustment is required in patients with mild, moderate or severe renal impairment not requiring dialysis. Sorafenib has not been studied in patients undergoing dialysis (see section Pharmacokinetic Properties – Renal impairment). Monitoring of fluid balance and electrolytes in patients at risk of renal dysfunction is advised.

4.3 Contra-indications

Hypersensitivity to the active substance or to any of the excipients listed in section 6.1.

4.4 Special warnings and precautions for use

Dermatological toxicities

Hand foot skin reaction (palmar-plantar erythrodysesthesia) and rash represent the most common adverse drug reactions with sorafenib. Rash and hand foot skin reaction are usually CTC (Common Toxicity Criteria) Grade 1 and 2 and generally appear during the first six weeks of treatment with sorafenib. Management of dermatological toxicities may include topical therapies for symptomatic relief, temporary treatment interruption and/or dose modification of sorafenib, or in severe or persistent cases, permanent discontinuation of sorafenib (see section 4.8).

Hypertension

An increased incidence of arterial hypertension was observed in sorafenib-treated patients. Hypertension was usually mild to moderate, occurred early in the course of treatment, and was amenable to management with standard antihypertensive therapy. Blood pressure should be monitored regularly and treated, if required, in accordance with standard medical practice. In cases of severe or persistent hypertension, or hypertensive crisis despite institution of antihypertensive therapy, permanent discontinuation of sorafenib should be considered (see section 4.8).

Aneurysms and artery dissections

The use of VEGF pathway inhibitors in patients with or without hypertension may promote the formation of aneurysms and/or artery dissections. Before initiating inFENIB, this risk should be carefully considered in patients with risk factors such as hypertension or history of aneurysm.

Hypoglycaemia

Decreases in blood glucose, in some cases clinically symptomatic and requiring hospitalization due to loss of consciousness, have been reported during sorafenib treatment. In case of symptomatic hypoglycaemia, sorafenib should be temporarily interrupted. Blood glucose levels in diabetic patients should be checked regularly in order to assess if anti-diabetic medicinal product's dosage needs to be adjusted.

Haemorrhage

An increased risk of bleeding may occur following sorafenib administration. If any bleeding event necessitates medical intervention, it is recommended that permanent discontinuation of sorafenib should be considered (see section 4.8).

Cardiac ischaemia and/or infarction

Temporary or permanent discontinuation of sorafenib should be considered in patients who develop cardiac ischaemia and/or infarction (see section 4.8).

QT interval prolongation

Sorafenib has been shown to prolong the QTc/QTc interval (see section 5.1), which may lead to an increased risk for ventricular arrhythmias. Use sorafenib with caution in patients who have, or may develop prolongation of QTc, such as patients with a congenital long QT syndrome, patients treated with a high cumulative dose of antiarrhythmic therapy, patients taking certain anti-arrhythmic medicines or other medicinal products that lead to QT prolongation, and those with electrolyte disturbances such as hypokalaemia, hypocalcaemia, or hypomagnesaemia. When using sorafenib in these patients, periodic monitoring with on-treatment electrocardiograms and electrolytes (magnesium, potassium, calcium) should be considered.

Gastrointestinal perforation

Gastrointestinal perforation is an uncommon event and has been reported in less than 1% of patients taking sorafenib. In some cases, this was not associated with apparent intra-abdominal tumour. Sorafenib therapy should be discontinued (see section 4.8).

Hepatic impairment

No data is available on patients with Child Pugh C (severe) hepatic impairment. Since sorafenib is mainly eliminated via the hepatic route exposure might be increased in patients with severe hepatic impairment (see sections 4.2).

Warfarin co-administration

Infrequent bleeding events or elevations in the International Normalised Ratio (INR) have been reported in some patients taking warfarin while on sorafenib therapy. Patients taking concomitant warfarin or phenprocoumon should be monitored regularly for changes in prothrombin time, INR or clinical bleeding episodes (see sections 4.5 and 4.8).

Wound healing complications

No formal studies of the effect of sorafenib on wound healing have been conducted. Temporary interruption of sorafenib therapy is recommended for precautionary reasons in patients undergoing major surgical procedures. There is limited clinical experience regarding the timing of reinitiation of therapy following major surgical intervention. Therefore, the decision to resume sorafenib therapy following a major surgical intervention should be based on clinical judgement of adequate wound healing.

Elderly population

Cases of renal failure have been reported. Monitoring of renal function should be considered.

Drug-drug interactions

Caution is recommended when administering sorafenib with compounds that are metabolised/eliminated predominantly by the UGT1A1 (e.g., irinotecan) or UGT1A9 pathways (see section 4.5).

Caution is recommended when sorafenib is co-administered with docetaxel (see section 4.5).

Co-administration of neomycin or other antibiotics that cause major ecological disturbances of the gastrointestinal microflora may lead to a decrease in sorafenib bioavailability (see section 4.5). The risk of reduced plasma concentrations of sorafenib should be considered before starting a treatment course with antibiotics.

Higher mortality has been reported in patients with squamous cell carcinoma of the lung treated with sorafenib in combination with platinum-based chemotherapies.

Disease specific warnings

Differentiated thyroid cancer (DTC)

Before initiating treatment, physicians are recommended to carefully evaluate the prognosis in the individual patient considering maximum lesion size, symptoms related to the disease and progression rate.

Management of suspected adverse drug reactions may require temporary interruption or dose reduction of sorafenib therapy.

Dose reductions were only partially successful in alleviating adverse reactions. Therefore, repeat evaluations of benefit and risk is recommended taking anti-tumour activity and tolerability into account.

Haemorrhage in DTC

Due to the potential risk of bleeding, tracheal, bronchial, and oesophageal infiltration should be treated with localized therapy prior to administering sorafenib in patients with DTC.

Hypocalcaemia in DTC

When using sorafenib in patients with DTC, close monitoring of blood calcium level is recommended. Severe hypocalcaemia should be corrected to prevent complications such as QT-prolongation or torsade de pointes (see section QT prolongation).

TSH suppression in DTC

Increases in TSH levels above 0.5mU/L were observed in sorafenib treated patients. When using sorafenib in DTC patients, close monitoring of TSH level is recommended.

Renal cell carcinoma

The benefit-risk of sorafenib in High-Risk Patients, according to MSKCC (Memorial Sloan Kettering Cancer Centre) prognostic group has not been evaluated.

4.5 Interaction with other medicinal products and other forms of interaction

Inhibitors of metabolic enzymes

Administration of rifampicin for 5 days before administration of a single dose of sorafenib resulted in an average 37 % reduction of sorafenib AUC. Other inducers of CYP3A4 activity and/or glucuronidation (e.g., Hypericum perforatum also known as St. John's wort, phenytoin, carbamazepine, phenobarbital, and dexamethasone) may also increase metabolism of sorafenib and thus decrease sorafenib concentrations.

CYP1A4 inhibitors

Ketocoazole, a potent inhibitor of CYP3A4, administered once daily for 7 days to healthy male volunteers did not alter the mean AUC of a single 50 mg dose of sorafenib. These data suggest that clinical pharmacokinetic interactions of sorafenib with CYP3A4 inhibitors are unlikely.

CYP2B6, CYP2C8 and CYP2C9 substrates

Sorafenib inhibited CYP2B6, CYP2C8 and CYP2C9 in vitro with similar potency. However, concomitant administration of sorafenib 400 mg twice daily with cyclophosphamide, a CYP2B6 substrate, or paclitaxel, a CYP2C8 substrate, did not result in a clinically meaningful inhibition. These data suggest that sorafenib at the recommended dose of 400 mg twice daily may not be an in vivo inhibitor of CYP2B6 or CYP2C8. Additionally, concomitant treatment with sorafenib and warfarin, a CYP2C9 substrate, did not result in changes in mean FII:NR compared to placebo. Thus, also the risk for a clinically relevant in vivo inhibition of CYP2C9 by sorafenib may be expected to be low. However, patients taking warfarin or phenprocoumon should have their INR checked regularly (see section 4.4).

CYP1A4, CYP2B6 and CYP2C9 substrates

Concomitant administration of sorafenib and midazolam, dextromethorphan or omeprazole, which are substrates for cytochromes CYP3A4, CYP2D6 and CYP2C19 respectively, did not alter the exposure of these agents. This indicates that sorafenib is neither an inhibitor nor an inducer of these cytochrome P450 isoenzymes. Therefore, clinical pharmacokinetic interactions of sorafenib with substrates of these enzymes are unlikely.

UGT1A1 and UGT1A9 substrates

In vitro, sorafenib inhibited glucuronidation via UGT1A1 and UGT1A9. The clinical relevance of this finding is unknown (see below and section 4.4).

In vitro studies of CYP enzyme induction

CYP1A2 and CYP3A4 activities were not altered after treatment of cultured human hepatocytes with sorafenib, indicating that sorafenib is unlikely to be an inducer of CYP1A2 and CYP3A4.

P-gp-substrates

In vitro, sorafenib has been shown to inhibit the transport protein p-glycoprotein (P-gp). Increased plasma concentrations of P-gp substrates such as digoxin cannot be excluded with concomitant treatment with sorafenib.

Combination with other anti-neoplastic agents

Sorafenib has been administered with a variety of other anti-neoplastic agents at their commonly used dosing regimens including gemtacinib, cisplatin, oxaliplatin, paclitaxel, carboplatin, capecitabine, doxorubicin, irinotecan, docetaxel and cyclophosphamide. Sorafenib had no clinically relevant effect on the pharmacokinetics of gemtacinib, cisplatin, carboplatin, oxaliplatin or cyclophosphamide.

Paclitaxel/carboplatin

Administration of paclitaxel (225 mg/m²) and carboplatin (AUC = 6) with sorafenib (≤ 400 mg twice daily), administered with a 3-day break in sorafenib dosing (two days prior to and on the day of paclitaxel/carboplatin administration), resulted in no significant effect on the pharmacokinetics of paclitaxel. Co-administration of paclitaxel (225 mg/m², once every 3 weeks) and carboplatin (AUC=6) with sorafenib (400 mg twice daily, without a break in sorafenib dosing) resulted in a 47% increase in sorafenib exposure, a 29% increase in paclitaxel exposure and a 50% increase in 6-OH paclitaxel exposure. The pharmacokinetics of carboplatin were unaffected.

These data indicate no need for dose adjustments when paclitaxel and carboplatin are co-administered with sorafenib with a 3-day break in sorafenib dosing (two days prior to and on the day of paclitaxel/carboplatin administration). The clinical significance of the increases in sorafenib and paclitaxel exposure, upon coadministration of sorafenib without a break in dosing, is unknown.

Capecitabine

Co-administration of capecitabine (750-1050 mg/m² twice daily, Days 1-14 every 21 days) and sorafenib (200 or 400 mg twice daily, continuous uninterrupted administration) resulted in no significant change in sorafenib exposure, but a 15–30% increase in capecitabine exposure and a 9–25% increase in 5-FU exposure. The clinical significance of these small to modest increases in capecitabine and 5-FU exposure when coadministered with sorafenib is unknown.

Doxorubicin/irinotecan

Concomitant treatment with sorafenib resulted in a 21 % increase in the AUC of doxorubicin. When administered with irinotecan, whose active metabolite SN-38 is further metabolised by the UGT1A1 pathway, there was a 67 ± 120 % increase in the AUC of SN-38 and a 26 ± 42 % increase in the AUC of irinotecan. The clinical significance of these findings is unknown (see section 4.4).

Docetaxel

Docetaxel (75 or 100 mg/m2 administered once every 21 days) when co-administered with sorafenib (200 mg twice daily or 400 mg twice daily administered on Days 2 through 19 of a 21-day cycle with a 3-day break in dosing around administration of docetaxel) resulted in a 36-80 % increase in docetaxel AUC and a 16-32 % increase in docetaxel C_{max}. Caution is recommended when sorafenib is co-administered with docetaxel (see section 4.4).

Combination with other agents

Neomycin

Co-administration of neomycin, a non-systemic antimicrobial agent used to eradicate gastrointestinal flora, interferes with the enterohepatic recycling of sorafenib (see section 5.2, Metabolism and Elimination), resulting in decreased sorafenib exposure. In healthy volunteers treated with a 5-day regimen of neomycin the average exposure to sorafenib decreased by 54%. Effects of other antibiotics have not been studied, but will likely depend on their ability to interfere with microorganisms with glucuronidase activity.

4.6 Pregnancy and lactation

Pregnancy

There are no data on the use of sorafenib in pregnant women. Studies in animals have shown reproductive toxicity including malformations. In rats, sorafenib and its metabolites were demonstrated to cross the placenta and sorafenib is anticipated to cause harmful effects on the foetus. Sorafenib should not be used during pregnancy unless clearly necessary, after careful consideration of the needs of the mother and the risk to the foetus. Women of childbearing potential must use effective contraception during treatment.

Lactation

It is not known whether sorafenib is excreted in human milk. In animals, sorafenib and/or its metabolites were excreted in milk. Because sorafenib could harm infant growth and development, women must not breast-feed during sorafenib treatment.

Fertility

Results from animal studies further indicate that sorafenib can impair male and female fertility.

4.7 Effects on ability to drive and use machines

No studies on the effects on the ability to drive and use machines have been performed. There is no evidence that sorafenib affects the ability to drive or to operate machinery.

4.8 Undesirable effects

The most important serious adverse reactions were myocardial infarction/ischaemia, gastrointestinal perforation, drug induced hepatitis, haemorrhage, and hypertension/hypertensive crisis.

The most common adverse reactions were diarrhoea, fatigue, alopecia, infection, hand foot skin reaction (corresponds to palmar plantar erythrodysesthesia syndrome in MedDRA) and rash.

Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness.

Table 1: All adverse reactions reported in patients in multiple clinical trials or through post-marketing use

System organ class	Very common	Common	Uncommon	Rare	Not known
Infections and infestations	infection	folliculitis			
Blood and lymphatic system disorders	lymphopenia	leucopenia neutropenia anemia thrombocytopenia			
Immune system disorders			Hypersensitivity reactions (including skin reactions and urticaria) anaphylactic reaction	angioedema	
Endocrine disorders		hypothyroidism	hyperthyroidism		
Metabolism and nutrition disorders	anorexia hypo-phosphataemia	hypocalcaemia hypokalaemia hypomagnesaemia hypoglycaemia	dehydration		
Psychiatric disorders		depression			
Nervous system disorders		peripheral sensory neuropathy dysgeusia	reversible posterior leukoencephalopathy*		encephalopathy*
Ear and labyrinth disorders		tinnitus			
Cardiac disorders		congestive heart failure* myocardial ischaemia and infarction*		QT prolongation	
Vascular disorders	haemorrhage (inc.gastrointestinal, respiratory tract* and cerebral haemorrhage*) hypertension	flushing	hypertensive crisis*		aneurysms and artery dissections
Respiratory, thoracic and mediastinal disorders		Rhinorrhoea dysphonia	interstitial lung disease-like events* (pneumonitis, radiation pneumonitis, acute respiratory distress, etc.)		
Gastrointestinal disorders	diarrhoea nausea vomiting constipation	stomatitis (including dry mouth and glossodynia) dyspepsia dysphagia gastro oesophageal reflux disease	pancreatitis gastritis gastrointestinal perforations*		
Hepatobiliary disorders			increase in bilirubin and jaundice, cholecystitis, cholangitis	drug induced hepatitis*	

Skin and subcutaneous tissue disorders	dry skin itch alopecia hand foot skin reaction** pruritus	keratocarcinoma squamous cell cancer of the skin dermatitis exfoliative acne skin desquamation hyperkeratosis	eczema erythema multiforme	radiation recall dermatitis Stevens-Johnson syndrome leucocytoclastic vasculitis toxic epidermal necrolysis*	
Musculoskeletal and connective tissue disorders	arthralgia	myalgia muscle spasms		rhabdomyolysis	
Renal and urinary disorders		renal failure proteinuria		nephrotic syndrome	
Reproductive system and breast disorders		erectile dysfunction	gynecomastia		
General disorders and administration site conditions	fatigue pain (including mouth, abdominal, bone, tumour pain and headache) fever	anemia influenza like illness mucosal inflammation			
Investigations	weight decreased increased amylase increased lipase	transient increase in transaminases	transient increase in blood alkaline phosphatase INR abnormal, prothrombin level abnormal		

* The adverse reactions may have a life-threatening or fatal outcome. Such events are either uncommon or less frequent than uncommon.

** Hand foot skin reaction corresponds to palmar plantar erythrodysesthesia syndrome in MedDRA.

* Cases have been reported in the post marketing setting.

Additional information on special populations

Certain adverse drug reactions such as hand foot skin reaction, diarrhoea, alopecia, weight decrease, hypertension, hypocalcaemia, and keratocarcinoma/squamous cell carcinoma of skin occurred at a substantially higher frequency in patients with differentiated thyroid compared to patients in the renal cell or hepatocellular carcinoma studies.

Laboratory test abnormalities in HCC and RCC patients

Increased lipase and amylase were very commonly reported.

Hypophosphataemia was a very common laboratory finding. The aetiology of hypophosphataemia associated with sorafenib is not known. CTCAE Grade 3 or 4 laboratory abnormalities included lymphopenia and neutropenia.

Hypocalcaemia occurring were low grade (CTCAE Grade 1 and 2). The aetiology of hypocalcaemia associated with sorafenib is not known.

Hypokalaemia occurring were low grade (CTCAE Grade 1). There were no reports of hypokalaemia CTCAE grade 4.

Laboratory test abnormalities in DTC patients

Hypocalcaemia occurring were low grade.

4.9 Overdose

There is no specific treatment for sorafenib overdose. The highest dose of sorafenib studied clinically is 800 mg twice daily. The adverse events observed at this dose were primarily diarrhoea and dermatological events. In the event of suspected overdose sorafenib should be withheld and supportive care instituted where necessary.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: Antineoplastic agents, protein kinase inhibitors, ATC code: L01EX02-Sorafenib. Sorafenib is a multit kinase inhibitor which has demonstrated both anti-proliferative and anti-angiogenic properties in vitro and in vivo.

Mechanism of action and pharmacodynamic effects

Sorafenib is a multit kinase inhibitor that decreases tumour cell proliferation in vitro. Sorafenib inhibits tumour growth of a broad spectrum of human tumour xenografts in athymic mice accompanied by a reduction of tumour angiogenesis.

Sorafenib inhibits the activity of targets present in the tumour cell (CRAF, BRAF, V600E BRAF, c-KIT, and FLT-3) and in the tumour vasculature (CRAF, VEGFR-2, VEGFR-3, and PDGFR- β). RAF kinases are serine/threonine kinases, whereas c-KIT, FLT-3, VEGFR-2, VEGFR-3, and PDGFR- β are receptor tyrosine kinases.

5.2 Pharmacokinetic properties

Absorption and distribution

After administration of sorafenib tablets the mean relative bioavailability is 38 - 49 % when compared to an oral solution. The absolute bioavailability is not known. Following oral administration sorafenib reaches peak plasma concentrations in approximately 3 hours. When given with a high-fat meal sorafenib absorption was reduced by 30 % compared to administration in the fasted state.

Mean C_{max} and AUC increased less than proportionally beyond doses of 400 mg administered twice daily. In vitro binding of sorafenib to human plasma proteins is 99.5 %.

Multiple dosing of sorafenib for 7 days resulted in a 2.5- to 7-fold accumulation compared to single dose administration. Steady state plasma sorafenib concentrations are achieved within 7 days, with a peak to trough ratio of mean concentrations of less than 2.

The steady-state concentrations of sorafenib administered at 400 mg twice daily were evaluated in DTC, RCC and HCC patients. The highest mean concentration was observed in DTC patients (approximately twice that observed in patients with RCC and HCC), though variability was high for all tumour types. The reason for the increased concentration in DTC patients is unknown.

Biotransformation and elimination

The elimination half-life of sorafenib is approximately 25 - 48 hours. Sorafenib is metabolised primarily in the liver and undergoes oxidative metabolism, mediated by CYP3A4, as well as glucuronidation mediated by UGT1A9. Sorafenib conjugates may be cleaved in the gastrointestinal tract by bacterial glucuronidase activity, allowing reabsorption of unconjugated active substance. Co-administration of neomycin has been shown to interfere with this process, decreasing the mean bioavailability of sorafenib by 54%.

Sorafenib accounts for approximately 70 - 85 % of the circulating analytes in plasma at steady state. Eight metabolites of sorafenib have been identified, of which five have been detected in plasma. The main circulating metabolite of sorafenib in plasma, the pyridine N-oxide, shows in vitro potency similar to that of sorafenib. This metabolite comprises approximately 9 - 16 % of circulating analytes at steady state. Following oral administration of a 100 mg dose of a solution formulation of sorafenib, 96 % of the dose was recovered within 14 days, with 77 % of the dose excreted in faeces, and 19 % of the dose excreted in urine as glucuronidated metabolites. Unchanged sorafenib, accounting for 51 % of the dose, was found in faeces but not in urine, indicating that biliary excretion of unchanged active substance might contribute to the elimination of sorafenib.

Pharmacokinetics in special populations

Analyses of demographic data suggest that there is no relationship between pharmacokinetics and age (up to 65 years), gender or body weight.

Pediatric population

No studies have been conducted to investigate the pharmacokinetics of sorafenib in paediatric patients.

Race

There are no clinically relevant differences in pharmacokinetics between Caucasian and Asian subjects.

Renal impairment

Steady state exposure to sorafenib was similar in patients with mild or moderate renal impairment compared to the exposures in patients with normal renal function. No relationship was observed between sorafenib exposure and renal function in subjects with normal renal function, mild, moderate or severe renal impairment. No data is available in patients' requiring dialysis.

Hepatic impairment

In hepatocellular carcinoma (HCC) patients with Child-Pugh A or B (mild to moderate) hepatic impairment, exposure values were comparable and within the range observed in patients without hepatic impairment. The pharmacokinetics (PK) of sorafenib in Child-Pugh A and B non-HCC patients were similar to the PK in healthy volunteers. There are no data for patients with Child-Pugh C (severe) hepatic impairment. Sorafenib is mainly eliminated via the liver, and exposure might be increased in this patient population.

5.3 Preclinical safety data

Not applicable

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Core: Hypromellose, croscarmellose sodium, microcrystalline cellulose, magnesium stearate, sodium laurylsulfate. Tablet coating: Hypromellose, titanium dioxide, macrogol, red iron oxide.

6.2 Incompatibilities

Not applicable.

6.3 Shelf-life

Aluminium-OPA/Ala/PVC Blisters - Please refer to expiry on the outer carton and blister. Aluminium-PVC/PE/PVDC Blisters - Please refer to expiry on the outer carton and blister.

6.4 Special precautions for storage

Do not store above 30°C

6.5 Nature and contents of container

inFENB Tablets 200 mg is available in cartons containing blisters of 6 x 10s. The blisters are available in Aluminium-OPA/Ala/PVC and Aluminium-PVC/PE/PVDC.

7. Product Registration Holder in Malaysia

Intega Sdn Bhd
8-08, Level 8, Menara MBMR,
No. 1, Jalan Syed Putra,
58000 Kuala Lumpur
Wilayah Persekutuan Kuala Lumpur

8. Name and Country of Manufacturer

PharOS MT Ltd.
HF02X, Hal Far Industrial Estate
Birzebbugia BBG3000
Malta

Date of revision of text

20 February 2025

PLF07-16100



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These panels shall be visible when folded

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